MUHAMMAD ZHAFARUL MAAHIY

Padang, Sumatera Barat, Indonesia | +62 857-6204-5359 | mzhafarulmaahiy@gmail.com
| Linkedln: www.linkedin.com/in/zhafarul-maahiy | GitHub: github.com/glifordosade

PROFESIONAL SUMMARY

Bachelor of Informatics graduate from Andalas University with a strong interest in Data Science and Artificial Intelligence. Possesses hands-on experience in Deep Learning research, particularly in developing an Indonesian emotional voice cloning model using the YourTTS architecture. Experienced in applying analytical and problem-solving approaches to research, working with data-driven models, and translating complex requirements into practical solutions. Also has professional experience in software development, backend systems, API integration, and database management, providing a solid technical foundation for data handling and information systems. Demonstrates strong analytical thinking, attention to detail, adaptability, and the ability to quickly learn new technologies and methodologies.

EDUCATION

Andalas University – Padang, Indonesia | Bachelor of Informatics (2022–2026) GPA:
3.78/4.00 (Cumlaude)
Undergraduate Thesis: "Development Of An Indonesian Emotional Voice Cloning Model Base On The Yourtts Architecture.

TECHNICAL SKILLS

- **Programming Languages**: Python, C#, Go, Java, Rust, PHP, Javascript, Node.js, TypeScript.
- **Database** : PostgreSQL, MySQL, MongoDB
- **Data Science & AI**: Data Processing, Data Validation, Machine Learning, Deep learning.
- **Frameworks & Libraries**: Pandas, Numpy, Solar, Seaborn, Scikit-learn, Pytorch, TailwindCSS, Reactjs, Vuejs, Nestjs, ReactNative, Flutter, Laravel, Next.js, Astrojs, Electron, Tauri, Gin Go Frameworks.
- **Software & Tools**: Microsoft Excel, Jupyter Notebook, Google Colab, VSCode, Docker, Git, GitHub, GitLab, Figma, Canva, Affinity and Google Workspace

PROFESSIONAL EXPERIENCE

PT.Imani Prima (Product and Business) – Jakarta Selatan, Indonesia
Software Developer (Intern) | July 2025 – August 2025

- Developed and maintained the landing page for Monstrack, ensuring a responsive, user-friendly, and visually appealing interface.
- Built the web application using Laravel and PostgreSQL, implementing backend functionality and database integration.
- Designed and developed an administrative dashboard using Filament to simplify content management and administrative operations.
- Created responsive and modern user interfaces using Tailwind CSS, delivering a consistent user experience across multiple devices.
- Implemented CRUD functionality and database interactions to support content management and website administration.
- Collaborated with the product and business team to gather requirements, translate business needs into technical solutions, and deliver features aligned with project objectives.
- Participated in debugging, testing, and optimizing the application to improve functionality, performance, and maintainability.

West Sumatra Provincial Government – Padang, Indonesia
Software Developer Project Manager (Contract) | February 2024 – October 2024

- Developed SIPPAK (Sistem Informasi Pengelolaan dan Pelayanan Administrasi Kepegawaian), a web-based application designed to digitize and streamline employee administration processes within the West Sumatra Provincial Government Secretariat.
- Developed core personnel administration modules for employee promotion, leave management, retirement processing, and official correspondence.
- Built the application using Laravel and MySQL, implementing backend functionality and database integration to support personnel data management.
- Designed and implemented database structures to manage employee records, administrative documents, and personnel service processes.
- Translated administrative workflows and business requirements into functional software features in collaboration with government staff and stakeholders.
- Implemented CRUD operations and data validation to improve the accuracy and consistency of personnel

administration records.
- Conducted application testing, debugging, and maintenance to ensure system reliability and usability for administrative staff.
- Contributed to the digitalization of manual administrative processes by providing a centralized platform for managing and accessing employee-related information.

PROJECTS

Emotional Indonesian Voice Cloning Using YourTTS By VITS Architecture | Undergraduate Thesis (November 2025 – May 2026)

- Developed an Indonesian emotional voice cloning model capable of generating speech from a target speaker.
- Develope a voice cloning model based on the YourTTS architecture, an extension of the VITS (Variational Inference with Adversarial Learning for End-to-End Text-to-Speech) architecture.
- Performed audio data preprocessing, including audio cleaning, segmentation, normalization, and format standardization to prepare speech datasets for model training.
- Processed and organized Indonesian speech datasets by preparing audio samples and corresponding text transcriptions for model training.
- Conducted audio preprocessing to ensure dataset consistency, including sampling rate standardization, audio duration filtering, and waveform normalization.
- Prepared and structured speaker audio data and text transcriptions to support the YourTTS training and voice cloning pipeline.
- Extended the voice cloning pipeline with emotion conditioning to control the emotional characteristics of generated speech across five categories: neutral, happy, disgust, surprise and disappointment.
- Trained and evaluated the speech synthesis model using Google Colab and GPU-based computational resources.
- Developed a FastAPI backend to expose the trained voice cloning model through API endpoints for application integration.
- Built a Next.js-based web interface to allow users to provide speaker audio, input text, and desired emotion and generate Indonesian speech through the trained model.
- Integrated the deep learning inference pipeline with the web application to provide an end-to-end voice cloning workflow from user input to generated audio.
- Conducted model testing and evaluation to assess the quality and emotional characteristics of the generated Indonesian speech.eaker's voice, input text and selected emotional expression..

BUMNAG Sepakat Matua Mudiak Landing Page | KKN Andalas University (January 2025 – February 2025)

- Developed a responsive landing page for BUMNag Matua Mudiak Sakato to strengthen the organization's online presence and improve public access to information.
- Built the web application using Laravel and PostgreSQL, implementing backend functionality and database integration for content management.
- Designed and developed a responsive user interface using Tailwind CSS, ensuring an intuitive user experience across desktop and mobile devices.
- Implemented features to showcase the village-owned enterprise's business units, products, and services through a structured and user-friendly interface.
- Developed an article management feature to publish and display news, updates, and information related to BUMNag activities.

E-Commerce Web Application (Lazada-Inspired Website) | Web Programming Course

- Developed a Lazada-inspired e-commerce web application using HTML, CSS, and PHP, implementing core online shopping functionalities.
- Developed a PHP-based backend and API to handle application logic and communication between the frontend and database.
- Designed and managed a MySQL database for product, user, and transaction-related data.
- Collaborated with team members in frontend development, backend integration dan database management.

Pimol Car rent appliaction | Mobile Application Course

- Developed a mobile-based car rental application using Android Studio and Java, implementing core features for browsing and renting available vehicles.
- Designed the application's UI/UX in Figma to create an intuitive and user-friendly mobile experience.
- Developed and integrated a PHP-based REST API to enable communication between the Android application and the backend server.
- Designed and managed the application's MySQL database to store and manage vehicle, user, and rental-related data.
- Integrated the Android application with backend services to retrieve, submit, and manage data through API requests.
- Collaborated with team members throughout the development lifecycle, from UI/UX design and implementation

to backend integration.

ORGANIZATION EXPERIENCE

HMIF 2023 Andalas University | Information and Media Member (June 2023 – August 2024)

- Managed and distributed organizational information through social media and digital communication channels.
- Created and published digital content to promote organizational activities, events, announcements and student programs.
- Designed visual content and promotional materials for social media and organizational campaigns.
- Collaborated with other divisions to gather information and ensure accurate and timely publication of organizational content.
- Documented and published information related to academic, organizational and student activities.
- Assisted in managing and organizing digital media assets, including images, videos, and other promotional materials.

Certifications and Training

Database Programming with SQL – Oracle Academy (July 2023)

Database Design – Oracle Academy (April 2023)

Fullstack Web Course – CodePolitan